SIADS 696 Milestone II Project Report
Team 18 - NHL Corsi Player Model

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Introduction

Hockey games are unpredictable at the single-game level, but certain performance indicators, especially puck possession at even strength, are more stable over time. This project asks whether team success can be better understood by combining repeatable possession metrics with structured information about player roles.

Our goal is twofold. First, we use unsupervised learning to group players into archetypes based on their usage, offensive production, defensive activity, and possession statistics. Second, we test whether team-level composition of these archetypes, together with rolling possession differentials, helps predict win probability in a supervised modeling framework.

We apply clustering methods to player-season data to identify three forward archetypes, and we build a logistic regression model to estimate team win probability using rolling even-strength metrics and time-on-ice–weighted archetype shares. Unlike many hockey analytics studies that focus only on possession or scoring models, our approach connects player role structure directly to team-level outcome prediction.

Our main findings are that three interpretable forward archetypes emerge from the data, although roles exist on a continuum rather than as sharply separated groups. In the supervised analysis, possession differential remains the strongest predictor of winning, but lineup composition provides additional predictive value. Overall, team structure matters, but territorial performance remains the dominant driver of results.

Related Work

One relevant study is *Using Machine Learning to Predict Future Points in the NHL* by Takehiro Matsuzawa (2017). This thesis uses machine learning methods and historical player statistics to forecast future point production for NHL players, illustrating how data-driven models can be applied to hockey performance prediction.

Another example is *Predictive Modeling and Analysis of Hockey Using Markov Chains* (Harsy et al., 2023), which uses a Markov process to model game states and estimate winning probabilities based on evolving game conditions. This work shows how state-based models can be useful for predicting game outcomes.

A further study, *A Bayesian Approach to Predict the Number of Goals in Hockey* (Sadeghkhani & Ahmed, 2019), applies Bayesian predictive models to estimate the distribution of goals scored in games, demonstrating the utility of probabilistic models in hockey analytics.

These studies show that machine learning, Bayesian methods, and Markov models have been applied to hockey analytics. Our project is different because it combines all three approaches in a single framework, from identifying player archetypes to simulating games and evaluating player and team performance.

Data Sources

Data were obtained from:

- NHL Public API – play-by-play events, player shifts, game results
- Spotrac.com – player salary and contract information
- Public statistical references – used to validate derived possession metrics

The NHL API returned data in JSON format, while salary information was retrieved via structured web scraping (HTML). All raw data were parsed and stored in a private SQL database before being transformed into modeling datasets. Final modeling tables were exported as Parquet files for analysis in Python.

Time Coverage

The database covers NHL regular seasons from 2015–2016 through 2024–2025, with the primary modeling window restricted to 2018–2019 through 2024–2025.

Key Raw Data Components

The main raw data elements include:

- Event-level play-by-play data
  (game ID, event type, timestamp, team, strength state)
- Player shift data
  (game ID, player ID, shift start/end, position)
- Derived possession metrics
  (Corsi For, Corsi Against, CF60, CA60, CF%)
- Player salary and biographical data

Modeling Datasets

From the raw data, three structured datasets were constructed:

1. Team–Game Dataset (Supervised Learning)
   16,938 team-game observations (2018–2025), including:
   - Rolling possession differentials (CF%, CF60, CA60)
   - Goals for/against
   - Win indicator
   - Time-on-ice–weighted archetype shares
2. Player–Game Dataset
   ~304,000 player-game records used to construct team aggregates.
3. Player–Season Dataset (Unsupervised Learning)
   ~3,500 forward player-seasons (2018–2025) with usage, offensive, defensive, and possession metrics used for clustering.

Data Scale

Across seven modeled seasons, the dataset includes:

- ~17,000 team-game observations
- ~300,000 player-game records
- ~3,500 player-season observations for clustering

Feature Engineering

Raw inputs consisted of NHL play-by-play events, player shift data, and salary information. From these, we constructed even-strength (5v5) possession metrics and time-on-ice statistics at the player-game level. Corsi For (CF) and Corsi Against (CA) were derived from shot attempts, and rate statistics (per 60 minutes) were computed.

For supervised modeling, player-game data were aggregated to the team-game level. Team and opponent metrics were paired within each game to create differential features (team − opponent). To prevent lookahead bias, all process variables were transformed into lagged 15-game rolling averages within each season.

For unsupervised learning, player-game statistics were aggregated to the player-season level. Features capture usage, offensive production, physical/defensive rates, faceoffs, and possession impact. Missing faceoff percentages were imputed to 0.50, low-quality observations were removed, and all clustering variables were standardized.

Final Unsupervised Features (15):
toi_per_game, es_share, g60, a60, p60, s60, hit60, blk60, take60, give60, penl60, fo_win_pct, cf60, ca60, cf_percent.

Final Supervised Features:
Rolling differential possession metrics (e.g., diff_cf_percent, diff_cf60, diff_ca60) and rolling time-on-ice–weighted archetype share differentials for forwards and defense.

Supervised Learning

Professional hockey outcomes are widely recognized as highly stochastic at the single-game level. While territorial advantage and puck possession are more repeatable than goal scoring, the final result of any given game is influenced by finishing variance, goaltending performance, special teams, and situational randomness. This presents a central modeling challenge: whether repeatable even-strength process metrics, augmented by structured roster composition information, contain sufficient signal to meaningfully estimate pre-game win probability.

The supervised objective of this project was to model team-level win probability using rolling even-strength performance metrics and lineup archetype composition. Rather than optimizing for deterministic winner selection, the analysis emphasizes probabilistic forecasting. In applied settings such as roster construction, matchup evaluation, and scenario analysis, calibrated probability estimates are more informative than raw classification accuracy.

Methods

The dataset used for supervised modeling consists of 16,938 team-game observations spanning seven regular seasons (2018–2019 through 2024–2025). Each NHL game contributes two observations, one for each team, making grouped validation essential to prevent leakage across train and validation folds. All modeling procedures employed GroupKFold cross-validation, grouping by game identifier to ensure that both rows corresponding to a single game remained within the same fold.

To eliminate lookahead bias, all process metrics were transformed into lagged rolling averages computed over the previous fifteen games. Specifically, for each team-season, base even-strength metrics were shifted by one game and then averaged using a rolling window. This ensured that no information from the

current game influenced the predictors. The primary engineered features consisted of possession-based differential metrics, including Corsi For percentage (CF%), Corsi For per 60 (CF60), and Corsi Against per 60 (CA60), each expressed as team-minus-opponent values to capture relative strength entering the matchup.

In addition to possession differentials, roster composition was incorporated through player archetype clusters split by forwards and defense. Player-season archetype labels were merged into player-game records, and team-level archetype shares were computed as time-on-ice–weighted proportions. Because archetypes were defined for regular players rather than all skaters, total labeled mass could sum to less than one; this nuance was preserved throughout model estimation and counterfactual simulation analyses.

The final supervised feature space consisted of 21 predictors: fifteen rolling process metrics and six rolling archetype share variables. Three model families were evaluated to test whether nonlinear structure improved predictive performance: L2-regularized logistic regression, random forest, and histogram gradient boosting. Hyperparameters were tuned using grid search within the grouped cross-validation framework. For logistic regression, regularization strength was varied across a logarithmic grid of inverse penalty values. For Random Forest, the number of estimators and maximum tree depth were explored. For histogram gradient boosting, learning rate, maximum tree depth, and minimum leaf size were tuned. All tuning was performed using five-fold GroupKFold validation to prevent leakage across paired team-game observations. Models were compared using ROC AUC as the primary discrimination metric, with Brier score and log loss used to assess probability quality.

Overall Results

Across five grouped cross-validation folds, logistic regression achieved a mean ROC AUC of 0.5855 ± 0.0045. Random forest produced nearly identical performance (0.5853 ± 0.0051), while histogram gradient boosting yielded slightly lower mean AUC (0.5826 ± 0.0075) and higher variance. Fold-level variability was small across models, suggesting stable generalization within the mixed-season cross-validation framework.

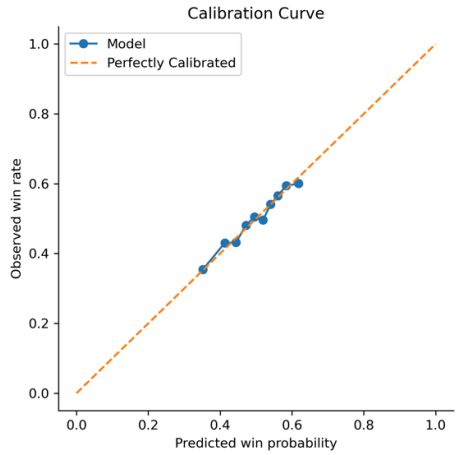
Table 1. Cross-Validated Model Performance

Model	**Mean ROC AUC**
Logistic Regression	0.58549 +/- 0.00454
Random Forest	0.58530 +/- 0.00507
Histogram Gradient Boosting	0.58263 +/- 0.00749

As shown in Table 1, logistic regression and random forest achieved nearly identical cross-validated performance, while histogram gradient boosting produced slightly lower mean AUC and higher variance. These minimal differences indicate that additional nonlinear capacity does not materially improve predictive discrimination within the engineered feature space. The dominant predictive signal appears largely linear in the differential possession and composition features.

To evaluate forward-looking robustness, a temporal holdout experiment was conducted in which models were trained on seasons through 2023–2024 and evaluated on 2024–2025. Performance declined to an AUC of approximately 0.564. This decline illustrates temporal concept drift: roster changes, strategic shifts, and league-wide stylistic evolution alter the mapping between features and outcomes. Mixed-season cross-validation therefore provides a moderately optimistic estimate of deployment performance, while the temporal holdout yields a more conservative benchmark.

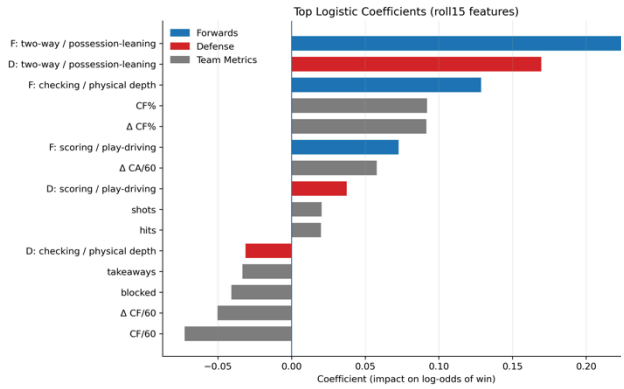
Calibration Analysis



The quality of predicted probabilities was assessed through calibration analysis using out-of-fold predictions. The calibration curve demonstrates that predicted probabilities track observed win rates reasonably well across most of the distribution. Mid-range probability bins align closely with empirical outcomes, while slight divergence appears in the tails, likely due to smaller sample sizes. Calibration quality is particularly important in this context because the primary use case is probabilistic forecasting rather than hard classification. Logistic regression's inherent probability structure contributes to stable calibration behavior.

Feature Ablation Analysis

Model interpretability was examined through analysis of the largest-magnitude logistic regression coefficients. The largest-magnitude coefficients correspond to forward and defense archetypes characterized as two-way, possession-leaning players. Specifically, Forward Archetype 0 (two-way/possession-leaning) and Defense Archetype 0 (two-way/possession-leaning) occupy the top two positions in the coefficient ranking, followed by a forward archetype characterized as checking or physical depth. Traditional process metrics such as CF% and differential CF% appear slightly lower in magnitude but remain among the most influential predictors.



This ordering is theoretically coherent. Possession-based metrics such as CF% capture recent realized territorial outcomes over a rolling window. By contrast, archetype shares encode the structural composition of the roster—namely, the proportion of players whose stylistic profiles emphasize two-way play and puck control. That these archetypes rank at the top of the coefficient distribution suggests that sustained deployment of territorial-strength skaters is strongly associated with win probability, even after controlling for recent possession metrics

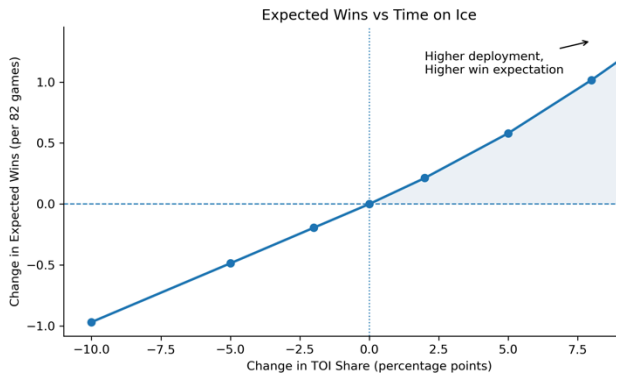
Importantly, this does not imply that archetype deployment is causally superior to possession performance. Logistic regression coefficients reflect conditional associations within a correlated feature space. Because possession metrics and possession-leaning archetype shares are naturally correlated, explanatory weight may be distributed across both structural and realized indicators of territorial control. The prominence of these archetype variables therefore suggests that roster composition captures aspects of sustained team quality that are not fully summarized by short rolling possession windows.

To formally assess the contribution of archetype composition, an ablation analysis was conducted in which all archetype share features were removed and the logistic model retrained. Cross-validated AUC

declined from approximately 0.585 to approximately 0.565, a decrease of roughly 0.02. This reduction exceeds fold-level noise and indicates that lineup composition provides independent predictive value beyond possession metrics alone. However, their magnitude underscores that composition alone does not overwhelm possession-based structure.

Sensitivity Analysis

Beyond static feature contribution, sensitivity analysis was conducted to quantify how changes in lineup composition affect predicted outcomes. Using the final logistic model (held fixed), Forward Archetype 0 (Two-way/possession leaning players) share was systematically perturbed while preserving total labeled time-on-ice mass.



The resulting curve exhibits an approximately linear relationship across ±10% adjustments. A five-percentage-point increase in Forward_0 deployment corresponds to roughly 0.62 additional expected wins over an 82-game season, while a ten-percentage-point increase corresponds to approximately 1.34 wins. Symmetric declines occur for negative adjustments. These effects, while modest, are directionally consistent and demonstrate that lineup structure can shift expected outcomes in quantifiable ways.

However, their magnitude underscores that composition alone cannot override possession trends. Territorial metrics remain the dominant predictors.

Tradeoffs

The supervised results reveal several important tradeoffs inherent to hockey outcome modeling. First, there is a tradeoff between model complexity and interpretability. While tree-based ensemble methods introduce nonlinear interactions, they did not meaningfully outperform logistic regression. Given comparable discrimination and slightly better calibration, logistic regression offers greater transparency without sacrificing predictive accuracy.

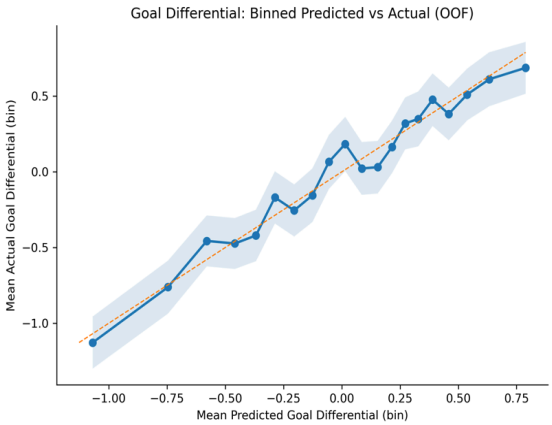
Second, there is a tradeoff between discrimination and realism. Mixed-season cross-validation yields higher AUC than temporal holdout evaluation, indicating that models trained on pooled seasons may overstate forward-looking performance due to concept drift. Temporal validation therefore provides a more conservative but realistic estimate of deploy-time robustness.

Third, there is a structural tradeoff between possession-based outcome proxies and margin prediction. While the rolling feature space meaningfully ranks win probability, it explains very little variance in goal differential. This highlights an intrinsic tradeoff between modeling directional competitive advantage and modeling exact scoring magnitude in a low-scoring, high-variance environment.

Finally, the sensitivity analysis underscores a tradeoff between lineup optimization and territorial dominance. While composition shifts produce measurable season-level changes in expected wins, their magnitude is bounded, suggesting that roster construction must operate within the structural constraints imposed by underlying possession dynamics.

Goal Differential Regression

To examine whether the same feature space could predict scoring margin, a regression model was fit to estimate goal differential.



Performance was substantially weaker, with an R^2 of approximately 0.03 and root mean squared error of approximately 2.58 goals. The binned predicted-versus-actual plot reveals substantial dispersion, indicating that margin prediction is dominated by variance in finishing and goaltending rather than by rolling possession metrics. These results reinforce that the engineered feature set is better suited for estimating relative win likelihood than for forecasting score magnitude. In a sport where a small number of goal events determine outcomes, predicting margin is intrinsically more volatile than predicting directional advantage

Failure Analysis

Exceptional Goaltending Performance
In several high-confidence loss predictions (e.g., predicted win probability > 0.70), the losing team faced unusually high opposing goaltender save percentages. In these games, possession metrics strongly favored the predicted winner, but finishing variance and elite goaltending suppressed scoring. This indicates the model does not capture game-level netminder performance.

Special Teams Effects
Some misclassifications occurred when teams with weaker even-strength possession recorded multiple power-play goals or allowed shorthanded goals. Because the feature space is restricted to even-strength metrics, special teams performance can override territorial advantage in individual games.

Low-Information Games Near 50% Probability
Games with predicted probabilities near 0.50 frequently resulted in misclassification due to inherent outcome uncertainty. These errors reflect stochastic variation rather than systematic bias in the feature space.
These failure modes suggest that incorporating goaltender statistics, special teams indicators, or expected goals metrics could improve predictive discrimination, particularly in high-confidence errors.

Unsupervised Learning

Player roles in professional hockey are multidimensional and cannot be adequately captured by a single performance metric. Forwards and defensemen differ substantially in usage patterns, time on ice distribution, and statistical profiles. To prevent positional differences from dominating the clustering structure, unsupervised learning was performed separately for forwards and defensemen.

The objective of the unsupervised component was to identify recurring player archetypes based on aggregated even-strength (5v5) performance profiles. Rather than imposing predefined role labels, clustering methods were used to allow stylistic player roles to emerge directly from the data.

Methods

Dataset and Feature Representation

The unsupervised dataset consists of player-season observations spanning the 2018–2019 through 2024–2025 NHL seasons. Each observation represents a single player-season aggregated at even strength (5v5).

Clustering was conducted separately for the two primary positional groups:

- Forwards (F)
- Defensemen (D)

For each position group, player-seasons were represented using 15 standardized performance features capturing several aspects of player behavior and on-ice impact.

Usage metrics

- time on ice per game (toi_per_game)
- even-strength share of ice time (es_share)

Offensive production rates

- goals per 60 minutes (g60)
- assists per 60 minutes (a60)
- points per 60 minutes (p60)
- shots per 60 minutes (s60)

Physical and defensive activity

- hits per 60 (hit60)
- blocked shots per 60 (blk60)
- takeaways per 60 (take60)
- giveaways per 60 (give60)
- penalties per 60 (penl60)

Faceoff performance

- faceoff win percentage (fo_win_pct)

Possession impact metrics

- Corsi for per 60 (cf60)
- Corsi against per 60 (ca60)
- Corsi percentage (cf_percent)

Missing faceoff percentages were imputed to 0.50, representing neutral performance for players without recorded faceoff attempts. All features were standardized to zero mean and unit variance prior to clustering to ensure that variables measured on different scales contributed equally to the clustering process.

Clustering Procedure

Clustering was performed separately for forwards and defensemen using two unsupervised learning approaches:

- **KMeans clustering**, a centroid-based partitioning algorithm that groups observations by minimizing within-cluster distance.
- **Gaussian Mixture Models (GMM)**, a probabilistic clustering method that models the data as a mixture of Gaussian distributions.

To maintain interpretability and consistency with commonly recognized hockey role structures, the number of clusters was fixed at **k = 3** for each positional group. This choice reflects the widely accepted conceptual grouping of player roles within each position (e.g., offensive, two-way, and defensive archetypes).

Although the number of clusters was fixed, clustering quality was evaluated using the silhouette score, which measures how well observations are separated from neighboring clusters relative to their internal cohesion.

Final Archetype Labels

The clustering process produces player-season archetype assignments.

Each observation receives:

- season
- player_id
- position group (F or D)
- cluster label

These labels represent data-driven player archetypes within each positional group.

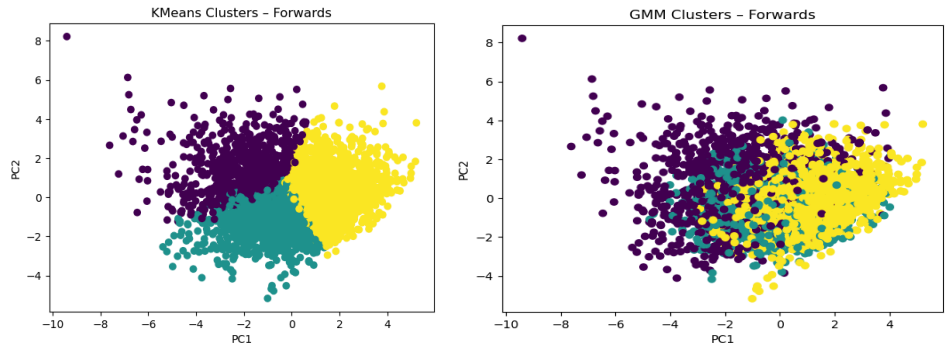
The resulting archetype assignments are later merged into team-game datasets to construct time-on-ice weighted roster composition features, which are used as inputs in the supervised win-probability modeling stage.

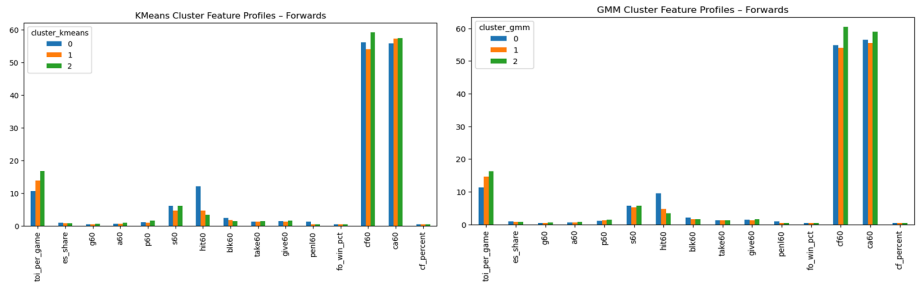
Results and Interpretation

Clustering quality was evaluated using silhouette scores for both KMeans and Gaussian Mixture Models within each positional group. The results are summarized below.

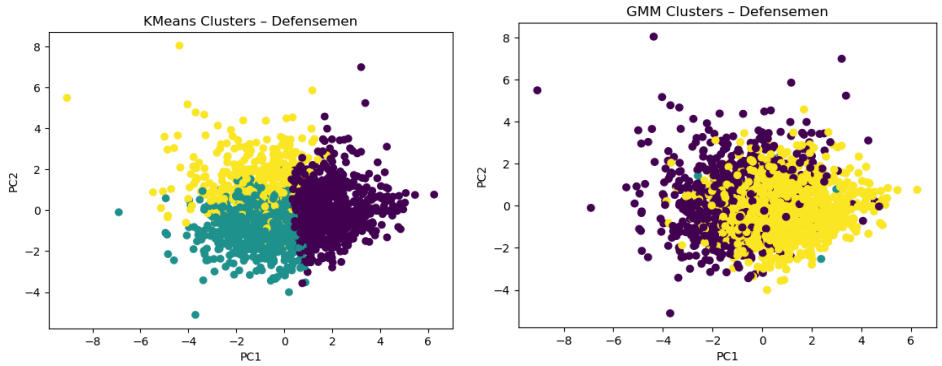
Position	Model	k	Silhouette
Forwards	KMeans	3	0.117
Forwards	GMM	3	0.036
Defensemen	KMeans	3	0.111
Defensemen	GMM	3	0.110

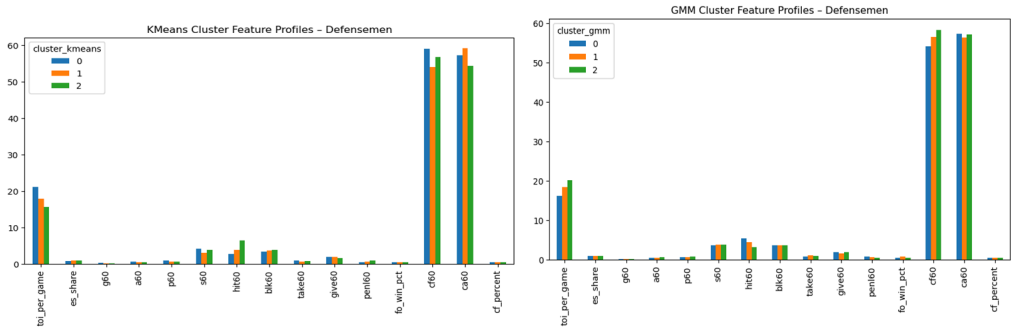
For forwards, KMeans produced substantially higher silhouette scores than the Gaussian Mixture Model, indicating stronger geometric separation between clusters.





For defensemen, both models produced similar silhouette values, suggesting that the cluster structure among defensemen is less sensitive to the clustering algorithm.





Overall, KMeans produced more stable and interpretable archetype groupings, particularly for forwards. Consequently, KMeans with k = 3 clusters per positional group was selected as the final clustering model.

The moderate silhouette values observed across both positional groups are consistent with the nature of sports performance data. Player roles in professional hockey rarely form sharply separated categories; instead, they exist along a continuum of playing styles, where players often blend characteristics of multiple archetypes.

Sensitivity Analysis

To assess the robustness of the clustering results, a sensitivity analysis was conducted by modifying the feature representation used in the final KMeans model.

The baseline clustering model used the full set of fifteen features, including physical and defensive activity metrics such as hits, blocked shots, takeaways, giveaways, and penalties. In the sensitivity experiment, these physical features were removed, leaving a reduced feature set consisting of usage, offensive production, faceoff performance, and possession metrics.

The clustering procedure was repeated using the same model specification (**KMeans with k = 3**) but with the reduced feature set. The resulting silhouette scores are summarized in the Table.

Position	Model	k	Silhouette
Forwards	KMeans (Full Features)	3	0.117
Forwards	KMeans (Reduced Features)	3	0.158
Defensemen	KMeans (Full Features)	3	0.111
Defensemen	KMeans (Reduced Features)	3	0.150

The reduced-feature specification produced higher silhouette scores for both positional groups, indicating stronger geometric separation between clusters when physical and defensive metrics are excluded.

This result suggests that usage, offensive production, and possession metrics provide clearer structural differentiation among player roles than when physical activity dimensions are included. Physical metrics introduce additional variability that may blur cluster boundaries in a purely geometric clustering framework.

Despite the improved silhouette scores under the reduced-feature specification, the full feature model was retained for the final archetype definition. Physical and defensive metrics represent important aspects of hockey playing style and contribute to more realistic and interpretable player archetypes. Retaining these variables ensures that the resulting clusters reflect meaningful stylistic differences rather than purely offensive production patterns.

Summary of Unsupervised Findings

- Clustering was performed separately for forwards and defensemen to account for structural differences in positional usage and statistical profiles.

- KMeans clustering with k = 3 per position group provided the most interpretable archetype structure.
- The final system defines six total player archetypes (three for forwards and three for defensemen).
- These archetypes are incorporated into later modeling stages as time-on-ice weighted roster composition features used in the supervised win-probability model.

Discussion

Supervised Learning

One notable outcome of the supervised analysis was the minimal performance gap between regularized logistic regression and more flexible tree-based ensemble models. Given the potential for nonlinear interactions between possession differentials and archetype shares, it was reasonable to anticipate that gradient boosting or random forests might extract additional predictive structure. Instead, the nearly identical ROC AUC scores across model families suggest that the dominant relationships within the engineered feature space are approximately linear, or that intrinsic outcome volatility limits the benefit of modeling nonlinear interactions.

Several methodological challenges arose during model development. A primary concern was preventing data leakage in a setting where each game contributes two team-level observations. This issue was addressed by implementing GroupKFold cross-validation grouped by game identifier, ensuring that paired observations from the same game were never split across training and validation folds.

Several extensions could strengthen the predictive framework. Incorporating goaltender performance indicators, expected goals models, and special teams metrics would likely improve discrimination, particularly in high-confidence misclassification cases. These variables would help capture contextual factors that materially influence single-game outcomes but were excluded from the present analysis.

The supervised analysis establishes an interpretable and well-calibrated probabilistic forecasting framework while also explicitly delineating the structural limits imposed by outcome volatility and missing contextual variables.

Unsupervised Learning

The unsupervised analysis reinforced that player roles in professional hockey are multidimensional and specific to position. An initial specification grouped all skaters into three archetypes, but this approach obscured meaningful structural differences between forwards and defensemen. Separating the clustering procedure by position and identifying three archetypes within each class, forwards and defensemen, produced clearer and more interpretable role distinctions, resulting in six total archetypes. This refinement illustrated that unsupervised learning results are highly sensitive to how the problem is framed and segmented. More broadly, the process emphasized that statistical separation alone is insufficient and must be evaluated alongside domain realism.

One notable outcome was the modest geometric separation of clusters even when the archetypes appeared coherent from a hockey perspective. Gaussian Mixture Models performed substantially worse than KMeans, suggesting that player performance profiles are not well represented by overlapping Gaussian distributions. The improvement in separation after dividing players by position was also informative, highlighting how combining fundamentally different role types can mask meaningful structural patterns.

Several challenges emerged during data preparation and model specification. A significant issue involved players who shared identical names across seasons or teams. Name-based aggregation risked conflating distinct individuals, because clustering operates at the player season level. This was resolved by relying exclusively on unique NHL player identifiers when constructing the player season dataset, reinforcing the importance of careful database design prior to unsupervised modeling.

Another challenge involved selecting the appropriate number of clusters. Silhouette scores did not identify a sharply optimal value, and early models that combined positions produced less interpretable structure. Incorporating domain knowledge proved essential. Clustering forwards and defensemen separately, with three archetypes each, balanced statistical evidence with realistic role differentiation.

With additional time and resources, the clustering framework could be extended by applying nonlinear dimensionality reduction techniques such as UMAP prior to clustering and conducting bootstrap based stability analysis to evaluate the robustness of assignments. Incorporating contextual deployment features such as zone starts, quality of competition, or teammate effects could generate more granular archetypes. Modeling role transitions across seasons would provide insight into how player identities evolve over time.

Ethical Considerations

Interpretation of Probabilistic Outputs

One ethical concern involves the misinterpretation of probabilistic model outputs as deterministic judgments about player or lineup value. Because the supervised model estimates win probability using rolling possession metrics and archetype composition, stakeholders could mistakenly interpret coefficient magnitudes or predicted probabilities as direct measures of individual player quality. In reality, the model captures conditional associations within a limited feature space and does not incorporate key factors such as goaltender performance, injuries, or special teams. To address this concern, it is important to clearly communicate that the model is intended as a decision-support tool rather than a definitive evaluator of player value, and to explicitly document its feature limitations, uncertainty, and calibration behavior.

Labeling and Role Classification

Potential misinterpretation of player archetypes as definitive or evaluative labels is an ethical concern for the unsupervised component. Because clustering imposes structure on multidimensional performance data, stakeholders may treat archetype assignments as fixed judgments about player ability rather than descriptive groupings based on statistical similarity. Cluster membership depends on feature selection, scaling choices, and model specification. Those interpretations could oversimplify player roles or influence decisions in ways not supported by the analysis.

To mitigate this risk, archetypes should be framed as exploratory summaries of playing style rather than measures of value, and their methodological sensitivity should be clearly documented. Emphasizing that roles exist on a continuum helps ensure the clustering results are used responsibly as decision-support tools rather than rigid classifications.

Statement of Work

Matthew Brady was responsible for the supervised learning component of the project. This included implementing the modeling pipeline, integrating player-season archetype shares into team-game data, and designing grouped cross-validation procedures to prevent leakage. He trained and evaluated multiple models, conducted sensitivity simulations, and failure analysis.

Eric Winiecke was the project creator and data engineer. He developed the repository, collected the raw datasets, and built the cleaned and aggregated tables used for analysis. He also led the unsupervised learning component, developing the player archetype clustering framework used to characterize player roles.

Evgeny Kuzmin contributed to the development of the supervised learning component by developing the rolling performance metrics and assisted with exploratory analysis and feature development used in the modeling process, and created visualizations used to interpret and present the unsupervised clustering results.